

Xobin Assessment Mega Practice Pack

This pack is designed based on your assessment pattern: 1. Quantitative Aptitude & Logical Reasoning 2. Verbal Ability & Reading Skills 3. SQL Programming 4. Short Programming (Java/C/Python/C++) 5. Data Structures & Algorithms

The questions below are similar to commonly asked placement and Xobin-style assessment questions.

SECTION 1: Quantitative Aptitude & Logical Reasoning (50 Questions)

Numbers, Percentages, Ratio, Profit & Loss

1. A shopkeeper gives 10% discount on ₹5000. Find selling price.
2. Ratio of A:B is 7:9. If A = 63, find B.
3. A number increased by 20% becomes 360. Find original number.
4. Simple interest on ₹8000 at 5% for 2 years?
5. Compound interest on ₹10000 at 10% for 2 years?
6. A train travels 240 km in 4 hours. Find speed.
7. Cost price = ₹500, selling price = ₹650. Find profit %.
8. If 25% of a number is 80, find the number.
9. Divide ₹5400 in ratio 2:3:4.
10. A can do work in 12 days, B in 18 days. Together?

Time & Work / Pipes

1. Pipe A fills tank in 20 min, Pipe B in 30 min. Together?
2. A alone completes work in 15 days. How much work in 1 day?
3. A and B together complete work in 6 days. If A alone takes 10 days, B alone?
4. Two pipes fill tank in 12 and 18 min. If outlet empties in 36 min, total time?
5. A works 4 days, B works 6 days. Total work done?

Time, Speed & Distance

1. A train 150 m long crosses pole in 10 sec. Speed?
2. Relative speed of trains moving opposite at 40 and 50 km/hr?
3. A cyclist travels 30 km at 15 km/hr. Time taken?
4. Boat speed in still water = 12 km/hr, stream = 3 km/hr. Downstream speed?
5. A car covers half distance at 40 km/hr and remaining at 60 km/hr. Average speed?

Logical Reasoning

1. Find next number: 2, 6, 12, 20, 30, ?
2. Find odd one: Apple, Mango, Carrot, Banana
3. Statement: All cats are animals. Some animals are black. Conclusion?
4. Mirror image of 2:35?
5. If SOUTH = 12345 and NORTH = 67895, what is SHORT?
6. Coding-decoding: CAT = DBU, DOG = ?
7. Blood relation: Ram is brother of Shyam's mother. Relation?
8. Clock angle at 3:30?
9. Find missing: 5, 10, 20, 40, ?
10. A is taller than B, B taller than C. Who shortest?

Probability & Permutation

1. Probability of getting head in coin toss?
2. Probability of getting even number on dice?
3. How many ways to arrange ABC?
4. Factorial of 5?
5. Number of diagonals in hexagon?

Data Interpretation Style

1. Company sales: Jan=100, Feb=120, Mar=150. % increase Jan to Mar?
2. Pie chart total students = 360. Science = 90°. Number?
3. Average of 10,20,30,40,50?
4. Median of 2,4,6,8,10?
5. Mode of 1,2,2,3,4?

Mixed Aptitude

1. HCF of 12 and 18?
 2. LCM of 8 and 12?
 3. Find missing term: 1,4,9,16,?
 4. If today is Monday, after 45 days?
 5. Area of circle radius 7?
 6. Perimeter of rectangle length 10 breadth 5?
 7. Convert 0.75 into fraction.
 8. Square root of 144?
 9. Cube of 12?
 10. Binary of decimal 10?
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SECTION 2: Verbal Ability & Reading Skills (50 Questions)

Grammar

1. Choose correct sentence.
2. Find synonym of "Rapid".
3. Find antonym of "Expand".
4. Fill in blank: She ___ going to school.
5. Identify tense: "He had completed work."
6. Active to passive conversion.
7. Direct to indirect speech.
8. Error spotting question.
9. Rearrange sentence logically.
10. Choose correct preposition.

Vocabulary

1. Synonym of "Brilliant".
2. Antonym of "Scarce".
3. Meaning of "Optimistic".
4. Idiom: "Break the ice".
5. One word substitution for "One who writes poems".
6. Correct spelling question.
7. Word analogy.
8. Sentence improvement.
9. Homophone question.
10. Prefix/suffix question.

Reading Comprehension

Passage-Based Questions:

1. Main idea of passage.
2. Tone of author.
3. Inference question.
4. Vocabulary from passage.
5. Suitable title.
6. Why did event happen?
7. What is implied?
8. Which statement true?
9. Which statement false?
10. Summarize paragraph.

Sentence Arrangement

1. Arrange sentences P,Q,R,S.
2. Find opening sentence.
3. Find concluding sentence.
4. Choose coherent paragraph.
5. Transition word question.

Communication Skills

1. Best reply in professional email.
2. Formal sentence selection.
3. Correct business communication.
4. Polite request sentence.
5. Choose concise sentence.

Mixed Verbal

1. Fill blank with article.
 2. Correct conjunction.
 3. Find adjective.
 4. Find adverb.
 5. Identify noun clause.
 6. Parallelism question.
 7. Subject-verb agreement.
 8. Conditional sentence.
 9. Punctuation correction.
 10. Choose grammatically correct sentence.
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SECTION 3: SQL Programming (50 Questions)

Assume table:

Employee(emp_id, name, dept, salary)

Basic Queries

1. Select all employees.
2. Select employee names only.
3. Find employees with salary > 50000.
4. Find employees from IT department.
5. Sort employees by salary descending.
6. Count total employees.
7. Find maximum salary.

8. Find minimum salary.
9. Find average salary.
10. Find employees whose name starts with 'A'.

WHERE, GROUP BY

1. Employees with salary between 30000 and 60000.
2. Count employees department-wise.
3. Find total salary department-wise.
4. Departments having more than 5 employees.
5. Employees not in HR.
6. Use IN operator.
7. Use LIKE operator.
8. Use DISTINCT keyword.
9. Find second highest salary.
10. Find duplicate records.

Joins

Tables: Department(dept_id, dept_name)

1. INNER JOIN example.
2. LEFT JOIN example.
3. RIGHT JOIN example.
4. FULL JOIN example.
5. Employees without department.
6. Find employee and department names.
7. Join using aliases.
8. Self join example.
9. Cross join example.
10. Natural join example.

Subqueries

1. Employees earning above average salary.
2. Highest salary in each department.
3. Third highest salary.
4. Employees in same dept as Rahul.
5. EXISTS query example.

Constraints & Functions

1. Create table with primary key.
2. Add foreign key.
3. NOT NULL example.
4. UNIQUE constraint example.
5. DEFAULT constraint example.

6. LENGTH() function.
7. UPPER() function.
8. LOWER() function.
9. ROUND() function.
10. NOW() function.

Advanced SQL

1. Delete duplicate rows.
 2. Update salary by 10%.
 3. Delete employees from HR.
 4. Create view.
 5. Difference between DELETE, DROP, TRUNCATE.
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SECTION 4: Short Programming (50 Coding Questions)

Arrays

1. Find largest element in array.
2. Find second largest element.
3. Reverse array.
4. Rotate array by k positions.
5. Find duplicates in array.
6. Remove duplicates.
7. Find missing number.
8. Merge two arrays.
9. Sort array ascending.
10. Kadane's algorithm maximum subarray sum.

Strings

1. Reverse string.
2. Check palindrome.
3. Count vowels.
4. Remove spaces.
5. Frequency of characters.
6. Check anagram.
7. Longest word in sentence.
8. First non-repeating character.
9. String compression.
10. Convert lowercase to uppercase.

Numbers

1. Prime number check.
2. Fibonacci series.
3. Factorial.
4. Armstrong number.
5. Perfect number.
6. GCD of two numbers.
7. LCM of two numbers.
8. Decimal to binary.
9. Binary to decimal.
10. Sum of digits.

Recursion

1. Recursive factorial.
2. Recursive Fibonacci.
3. Power function.
4. Tower of Hanoi.
5. Sum of array recursively.

Searching & Sorting

1. Binary search.
2. Linear search.
3. Bubble sort.
4. Selection sort.
5. Insertion sort.
6. Merge sort.
7. Quick sort.
8. Count inversions.
9. Find kth largest.
10. Two sum problem.

Patterns & Misc

1. Star pyramid pattern.
 2. Floyd's triangle.
 3. Matrix addition.
 4. Matrix multiplication.
 5. Transpose of matrix.
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SECTION 5: Data Structures & Algorithms MCQs (50 Questions)

Arrays & Strings

1. Time complexity of array access?
2. Best case of binary search?
3. Worst case of linear search?
4. Which sorting is stable?
5. Space complexity of merge sort?
6. Which DS uses contiguous memory?
7. Dynamic array example?
8. String immutability in Java?
9. Difference between array and linked list?
10. Time complexity of insertion at end in array?

Linked List

1. Advantage of linked list.
2. Circular linked list use.
3. Doubly linked list extra memory?
4. Delete node complexity.
5. Detect cycle algorithm.

Stack & Queue

1. Stack follows?
2. Queue follows?
3. Applications of stack.
4. Infix to postfix uses?
5. Queue implementation.
6. Priority queue meaning.
7. Deque meaning.
8. Overflow in stack.
9. Underflow in queue.
10. Circular queue advantage.

Trees

1. Root node meaning.
2. Height of tree.
3. Binary search tree property.
4. Inorder traversal.
5. Preorder traversal.
6. Postorder traversal.

7. AVL tree purpose.
8. Heap property.
9. Min heap root contains?
10. Full binary tree meaning.

Graphs

1. BFS uses which DS?
2. DFS uses which DS?
3. Dijkstra algorithm purpose.
4. Minimum spanning tree.
5. Cycle detection in graph.

Algorithms

1. Time complexity of quicksort average case.
2. Worst case of quicksort.
3. Greedy algorithm example.
4. Dynamic programming example.
5. Recursion base condition.
6. Big O notation meaning.
7. Best sorting algorithm average.
8. HashMap average search complexity.
9. Collision in hashing.
10. Difference between stack and heap memory.

IMPORTANT LAST-MINUTE TOPICS TO REVISE

Aptitude

- Percentages
- Ratio & Proportion
- Time & Work
- Pipes & Cisterns
- Profit & Loss
- Speed, Time & Distance
- Number Series

SQL

- SELECT
- WHERE
- GROUP BY
- JOINS
- SUBQUERIES

- Aggregate Functions

Coding

- Arrays
- Strings
- HashMap
- Prefix Sum
- Binary Search
- Sorting
- Sliding Window

DSA

- Time Complexity
- Stack vs Queue
- Tree Traversals
- BFS & DFS
- HashMap
- Sorting Complexities

MOST IMPORTANT CODING QUESTIONS TO PRACTICE TONIGHT

1. Two Sum
2. Maximum Subarray Sum
3. Reverse String
4. Palindrome
5. Fibonacci
6. Binary Search
7. Find Missing Number
8. Frequency Count using HashMap
9. Anagram Check
10. Prefix Sum Problems

INTERVIEW TIPS FOR TOMORROW

- Read question completely before coding.
- Don't spend too much time on one question.
- Start from easy coding questions.
- For MCQs eliminate wrong options first.
- In SQL use proper syntax carefully.

- Dry run your code with sample input.
- Handle edge cases.
- Stay calm even if one section goes difficult.

You already practiced aptitude, ratio, proportion, pipes, work & efficiency. That gives you a strong advantage. Focus mainly on:

- Coding speed
- SQL syntax
- DSA MCQs

All the best for your Xobin assessment 🚀